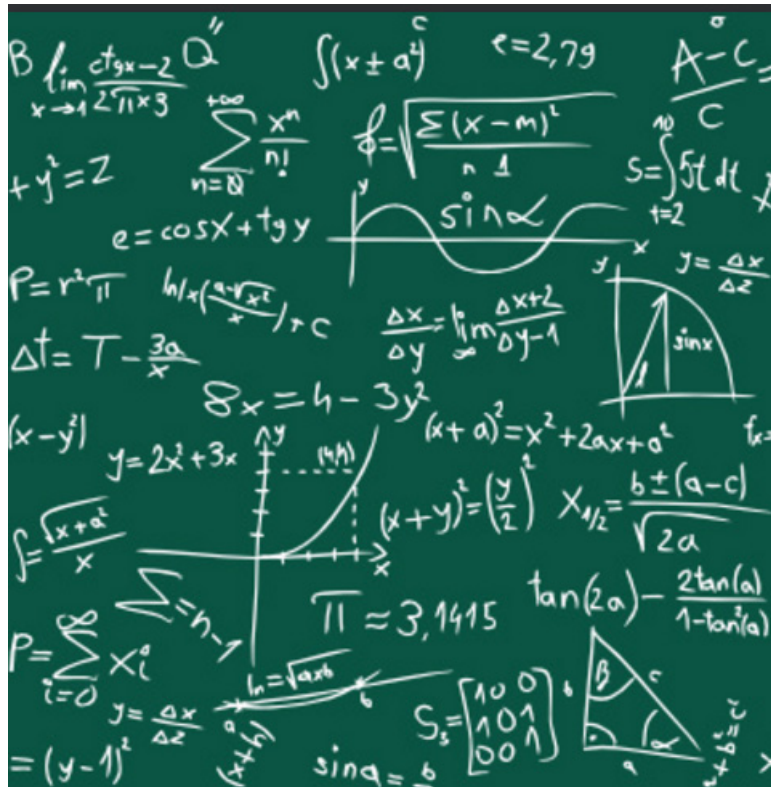


Exercises: Antiderivatives



- **Authors:** Fabio Furini and Gianluca Priori.
- **Book:** “Esercitazioni Analisi matematica 1”, Marco Bramanti, Esculapio.

Exercise 1

Compute the following antiderivatives on the intervals where they are defined

$$(a) \int \left(2x + \sqrt[3]{x} - \frac{2}{\sqrt{x}} \right) dx, \quad (b) \int \frac{x - x^4}{\sqrt{x}} dx$$

$$(c) \int \tan^2 x \, dx, \quad (d) \int (3 - x)^5 dx$$

$$(e) \int \frac{1}{2 - 3x} dx, \quad (f) \int x^2 e^{x^3+1} dx$$

$$(g) \int x \sqrt{x^2 + 1} dx, \quad (h) \int \frac{\sin 2x}{1 + \sin^2 x} dx$$

$$(i) \int \frac{e^x}{e^x + 3} dx, \quad (j) \int \frac{1}{\tan x} dx$$

(a)

$$\begin{aligned} \int \left(2x + \sqrt[3]{x} - \frac{2}{\sqrt{x}} \right) dx &= \int (2x + x^{1/3} - 2x^{-1/2}) dx \\ &= x^2 + \frac{3}{4}x^{4/3} - 4x^{1/2} + c = x^2 + \frac{3}{4}\sqrt[3]{x^4} - 4\sqrt{x} + c \end{aligned}$$

on any interval I such that $I \subset (0, +\infty)$.

(b)

$$\begin{aligned} \int \frac{x - x^4}{\sqrt{x}} dx &= \int (x^{1/2} - x^{7/2}) dx = \\ \frac{2}{3}x^{3/2} - \frac{2}{9}x^{9/2} + c &= \frac{2}{3}x\sqrt{x} - \frac{2}{9}x^4\sqrt{x} + c \end{aligned}$$

on any interval I such that $I \subset (0, +\infty)$.

(c)

$$\int \tan^2 x \, dx = \int [(1 + \tan^2 x) - 1] dx = \tan x - x + c$$

on any interval I that does not contain points of the form $\pi/2 + k\pi$ with $k \in \mathbb{Z}$.

(d)

$$\int (3-x)^5 dx = -\int (x-3)^5 dx = -\frac{1}{6}(x-3)^6 + c$$

on any interval I of $\mathbb{R}$.

(e)

$$\int \frac{1}{2-3x} dx = -\int \frac{1}{3x-2} dx = -\frac{1}{3} \log|3x-2| + c$$

on any interval I such that $2/3 \notin I$.

(f)

$$\int x^2 e^{x^3+1} dx = \frac{1}{3} e^{x^3+1} + c$$

on any interval I of $\mathbb{R}$.

(g)

$$\int x \sqrt{x^2+1} dx = \frac{1}{3}(x^2+1)^{3/2} + c = \frac{1}{3}(x^2+1) \sqrt{x^2+1} + c$$

on any interval I of $\mathbb{R}$.

(h)

$$\int \frac{\sin 2x}{1+\sin^2 x} dx = \int \frac{2 \sin x \cos x}{1+\sin^2 x} dx = \log(1+\sin^2 x) + c$$

on any interval I of $\mathbb{R}$.

(i)

$$\int \frac{e^x}{e^x + 3} dx = \log(e^x + 3) + c$$

on any interval I of $\mathbb{R}$.

(j)

$$\int \frac{1}{\tan x} dx = \int \frac{\cos x}{\sin x} dx = \log |\sin x| + c$$

on any interval I that does not contain points of the form $k\pi$ with $k \in \mathbb{Z}$.

Exercise 2

Compute the following antiderivatives using integration by parts

(a) $\int x \sin x \, dx$, (b) $\int x^2 \cos x \, dx$

(c) $\int \arcsin x \, dx$, (d) $\int \sin^2 x \, dx$

(a)

$$\int x \sin x \, dx = -x \cos x + \int \cos x \, dx = -x \cos x + \sin x + c$$

on any interval I of $\mathbb{R}$.

(b) Integrating by parts twice in a row, we have

$$\int x^2 \cos x \, dx = x^2 \sin x - 2 \int x \sin x \, dx =$$

$$x^2 \sin x + 2x \cos x - 2 \int \cos x \, dx = x^2 \sin x + 2x \cos x - 2 \sin x + c$$

on any interval I of $\mathbb{R}$.

(c) Taking the constant function 1 as the factor to be integrated, we have

$$\int \arcsin x \, dx = x \arcsin x - \int \frac{x}{\sqrt{1-x^2}} dx = x \arcsin x + \sqrt{1-x^2} + c$$

on any interval $I \subset (-1, 1)$.

(d) Integrating by parts and using $\cos^2 x = 1 - \sin^2 x$, we obtain

$$\begin{aligned} \int \sin^2 x \, dx &= -\sin x \cos x + \int \cos^2 x \, dx \\ &= -\sin x \cos x + \int 1 \, dx - \int \sin^2 x \, dx \end{aligned}$$

from which

$$\int \sin^2 x \, dx = -\sin x \cos x + x - \int \sin^2 x \, dx + c$$

hence, moving $\int \sin^2 x \, dx$ to the left-hand side,

$$2 \int \sin^2 x \, dx = -\sin x \cos x + x + c$$

and finally, still denoting by c the arbitrary constant $c/2$,

$$\int \sin^2 x \, dx = -\frac{1}{2} \sin x \cos x + \frac{1}{2} x + c$$

on any interval I of $\mathbb{R}$.

Exercise 3

Compute the following antiderivatives using integration by substitution

$$(a) \int \frac{e^x - 1}{e^x + 1} dx, \quad (b) \int \frac{\sqrt{x}}{1+x} dx, \quad (c) \int x\sqrt{x+1} dx$$

(a) Setting $y = e^x$, $y > 0$, $x \in \mathbb{R}$, we have $x = \log y$ and we write $dx = \frac{1}{y} dy$, hence

$$\int \frac{e^x - 1}{e^x + 1} dx = \int \frac{y - 1}{y(y + 1)} dy.$$

Using partial fractions, we obtain

$$\frac{y - 1}{y(y + 1)} = -\frac{1}{y} + \frac{2}{y + 1}$$

from which, taking into account that $y > 0$,

$$\int \frac{y - 1}{y(y + 1)} dy = -\log y + 2 \log(y + 1) + c.$$

It follows, substituting back $y = e^x$,

$$\int \frac{e^x - 1}{e^x + 1} dx = -\log e^x + 2 \log(e^x + 1) + c = -x + 2 \log(e^x + 1) + c$$

on any interval I of $\mathbb{R}$.

(b) Setting $y = \sqrt{x}$, $x, y > 0$, we have $x = y^2$ and we write $dx = 2y dy$, hence

$$\int \frac{\sqrt{x}}{1 + x} dx = 2 \int \frac{y^2}{1 + y^2} dy.$$

From

$$\frac{y^2}{y^2 + 1} = \frac{y^2 + 1 - 1}{y^2 + 1} = 1 - \frac{1}{y^2 + 1}$$

we then have

$$2 \int \frac{y^2}{1 + y^2} dy = 2y - 2 \arctan y + c.$$

It follows, substituting back $y = \sqrt{x}$,

$$\int \frac{\sqrt{x}}{1 + x} dx = 2\sqrt{x} - 2 \arctan \sqrt{x} + c$$

on any interval $I \subset (0, +\infty)$.

(c) Setting $y = \sqrt{x+1}$, $x > -1$, $y > 0$, we have $x = y^2 - 1$ and we write $dx = 2y \, dy$, hence

$$\int x\sqrt{x+1} \, dx = 2 \int (y^2 - 1)y^2 \, dy = 2 \int (y^4 - y^2) \, dy = \frac{2}{5}y^5 - \frac{2}{3}y^3 + c.$$

It follows, substituting back $y = \sqrt{x+1}$,

$$\int x\sqrt{x+1} \, dx = \frac{2}{5}(x+1)^{5/2} - \frac{2}{3}(x+1)^{3/2} + c$$

on any interval $I \subset (-1, +\infty)$.

Exercise 4

Compute the following antiderivatives of rational functions

(a) $\int \frac{5}{x^2 + 2x + 1} \, dx$, (b) $\int \frac{1}{x^2 + x + 1} \, dx$, (c) $\int \frac{x}{x^2 + x + 1} \, dx$, (d) $\int \frac{x^3 + 1}{x^2 - 3x + 2} \, dx$

(e) $\int \frac{x}{x^3 + 3x^2 + 3x + 1} \, dx$, (f) $\int \frac{1}{x^4 - 1} \, dx$, (g) $\int \frac{x+1}{x^3 - x^2} \, dx$, (h) $\int \frac{1}{x^3 + 1} \, dx$

(a) The given integral is immediate:

$$\int \frac{5}{x^2 + 2x + 1} \, dx = \int \frac{5}{(x+1)^2} \, dx = -\frac{5}{x+1} + c$$

on any interval I such that $-1 \notin I$.

(b) Writing the (irreducible) denominator as a sum of squares:

$$\int \frac{1}{x^2 + x + 1} \, dx = \int \frac{1}{(x+1/2)^2 + 3/4} \, dx = \frac{2}{\sqrt{3}} \arctan \left[\frac{2}{\sqrt{3}} \left(x + \frac{1}{2} \right) \right] + c$$

on any interval I of $\mathbb{R}$.

(c) Using also the previous exercise:

$$\int \frac{x}{x^2 + x + 1} dx = \frac{1}{2} \int \frac{2x + 1}{x^2 + x + 1} dx - \frac{1}{2} \int \frac{1}{x^2 + x + 1} dx =$$
$$\frac{1}{2} \log(x^2 + x + 1) - \frac{1}{\sqrt{3}} \arctan \left[\frac{2}{\sqrt{3}} \left(x + \frac{1}{2} \right) \right] + c$$

on any interval I of $\mathbb{R}$.

(d) We perform polynomial long division:

$$\frac{x^3 + 1}{x^2 - 3x + 2} = x + 3 + \frac{7x - 5}{x^2 - 3x + 2}.$$

Setting up the partial fractions

$$\frac{7x - 5}{x^2 - 3x + 2} = \frac{7x - 5}{(x - 1)(x - 2)} = \frac{A}{x - 1} + \frac{B}{x - 2},$$

we obtain

$$A = -2, B = 9.$$

In conclusion:

$$\int \frac{x^3 + 1}{x^2 - 3x + 2} dx = \int \left(x + 3 - \frac{2}{x - 1} + \frac{9}{x - 2} \right) dx =$$
$$\frac{1}{2} x^2 + 3x - 2 \log|x - 1| + 9 \log|x - 2| + c$$

on any interval I such that $1 \notin I, 2 \notin I$.

(e) We set up the partial fractions:

$$\frac{x}{x^3 + 3x^2 + 3x + 1} = \frac{x}{(x+1)^3} = \frac{A}{x+1} + \frac{B}{(x+1)^2} + \frac{C}{(x+1)^3}.$$

We obtain

$$A = 0, B = 1, C = -1.$$

In conclusion

$$\begin{aligned} \int \frac{x}{x^3 + 3x^2 + 3x + 1} dx &= \int \left(\frac{1}{(x+1)^2} - \frac{1}{(x+1)^3} \right) dx \\ &= -\frac{1}{x+1} + \frac{1}{2} \frac{1}{(x+1)^2} + c \end{aligned}$$

on any interval I such that $-1 \notin I$.

(f) We set up the partial fractions:

$$\frac{1}{x^4 - 1} = \frac{1}{(x-1)(x+1)(x^2+1)} = \frac{A}{x-1} + \frac{B}{x+1} + \frac{Cx+D}{x^2+1}.$$

We obtain

$$A = \frac{1}{4}, B = -\frac{1}{4}, C = 0, D = -\frac{1}{2}.$$

In conclusion

$$\begin{aligned} \int \frac{1}{x^4 - 1} dx &= \int \left(\frac{1}{4} \frac{1}{x-1} - \frac{1}{4} \frac{1}{x+1} - \frac{1}{2} \frac{1}{x^2+1} \right) dx \\ &= \frac{1}{4} \log|x-1| - \frac{1}{4} \log|x+1| - \frac{1}{2} \arctan x + c \end{aligned}$$

on any interval I such that $\pm 1 \notin I$.

(g) We set up the partial fractions:

$$\frac{x+1}{x^3-x^2} = \frac{x+1}{x^2(x-1)} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x-1}.$$

We obtain

$$A = -2, B = -1, C = 2.$$

In conclusion

$$\begin{aligned}\int \frac{x+1}{x^3-x^2} dx &= \int \left(-\frac{2}{x} - \frac{1}{x^2} + \frac{2}{x-1} \right) dx \\ &= -2\log|x| + \frac{1}{x} + 2\log|x-1| + c\end{aligned}$$

on any interval I such that $0, 1 \notin I$.

(h) We set up the partial fractions:

$$\frac{1}{x^3+1} = \frac{1}{(x+1)(x^2-x+1)} = \frac{A}{x+1} + \frac{Bx+C}{x^2-x+1}.$$

We obtain

$$A = \frac{1}{3}, B = -\frac{1}{3}, C = \frac{2}{3}.$$

Hence

$$\begin{aligned}\int \frac{1}{x^3+1} dx &= \int \left(\frac{1}{3} \frac{1}{x+1} + \frac{1}{3} \frac{-x+2}{x^2-x+1} \right) dx \\ &= \frac{1}{3} \log|x+1| - \frac{1}{6} \int \frac{2x-1}{x^2-x+1} dx + \frac{1}{2} \int \frac{1}{x^2-x+1} dx \\ &= \frac{1}{3} \log|x+1| - \frac{1}{6} \log(x^2-x+1) + \frac{1}{2} \int \frac{1}{(x-1/2)^2+3/4} dx \\ &= \frac{1}{3} \log|x+1| - \frac{1}{6} \log(x^2-x+1) + \frac{1}{\sqrt{3}} \arctan \left[\frac{2}{\sqrt{3}} \left(x - \frac{1}{2} \right) \right] + c\end{aligned}$$

on any interval I such that $-1 \notin I$.

Exercise 5

Compute the following antiderivatives

$$(a) \int \frac{e^x + 1}{e^{2x} + 1} dx, \quad (b) \int x^2 \log(x+1) dx, \quad (c) \int \frac{\tan x}{\sin^2 x - \cos^2 x} dx$$

$$(d) \int \frac{x + \sqrt{x}}{2 + \sqrt{x}} dx, \quad (e) \int \frac{1}{\cos x} dx, \quad (f) \int \sqrt{4 - x^2} dx$$

(a) With the substitution $y = e^x$, $y > 0$, $x \in \mathbb{R}$, we have $x = \log y$, we write $dx = \frac{1}{y} dy$, and we obtain

$$\int \frac{e^x + 1}{e^{2x} + 1} dx = \int \frac{y + 1}{y(y^2 + 1)} dy.$$

We set up the partial fractions:

$$\frac{y + 1}{y(y^2 + 1)} = \frac{A}{y} + \frac{By + C}{y^2 + 1}.$$

We have

$$A = 1, \quad B = -1, \quad C = 1,$$

from which, taking into account also that $y > 0$,

$$\begin{aligned} \int \frac{y + 1}{y(y^2 + 1)} dy &= \int \left(\frac{1}{y} - \frac{y}{y^2 + 1} + \frac{1}{y^2 + 1} \right) dy \\ &= \log y - \frac{1}{2} \log(y^2 + 1) + \arctan y + c. \end{aligned}$$

Then, substituting back $y = e^x$, we have

$$\int \frac{e^x + 1}{e^{2x} + 1} dx = x - \frac{1}{2} \log(e^{2x} + 1) + \arctan e^x + c$$

on any interval I of $\mathbb{R}$.

(b) Integrating by parts and then computing the integral of a rational function, we have

$$\begin{aligned}
 \int x^2 \log(x+1) dx &= \frac{1}{3} x^3 \log(x+1) - \frac{1}{3} \int \frac{x^3}{x+1} dx \\
 &= \frac{1}{3} x^3 \log(x+1) - \frac{1}{3} \int (x^2 - x + 1) dx + \frac{1}{3} \int \frac{1}{x+1} dx \\
 &= \frac{1}{3} x^3 \log(x+1) - \frac{1}{9} x^3 + \frac{1}{6} x^2 - \frac{1}{3} x + \frac{1}{3} \log(x+1) + c
 \end{aligned}$$

on any interval $I \subset (-1, +\infty)$.

(c) From $\tan x = \frac{\sin x}{\cos x}$ and $\sin^2 x = 1 - \cos^2 x$, we have

$$\int \frac{\tan x}{\sin^2 x - \cos^2 x} dx = \int \frac{\sin x}{\cos x(1 - 2\cos^2 x)} dx.$$

Now setting $y = \cos x$, we write $dy = -\sin x dx$ and we obtain

$$\int \frac{\sin x}{\cos x(1 - 2\cos^2 x)} dx = \frac{1}{2} \int \frac{1}{y(y - \sqrt{1/2})(y + \sqrt{1/2})} dy.$$

Using partial fractions, we then have

$$\frac{1}{y(y - \sqrt{1/2})(y + \sqrt{1/2})} = -\frac{2}{y} + \frac{1}{y - \sqrt{1/2}} + \frac{1}{y + \sqrt{1/2}},$$

from which

$$\frac{1}{2} \int \frac{1}{y(y - \sqrt{1/2})(y + \sqrt{1/2})} dy = -\log|y| + \frac{1}{2} \log|y^2 - 1/2| + c.$$

In conclusion,

$$\int \frac{\tan x}{\sin^2 x - \cos^2 x} dx = -\log|\cos x| + \frac{1}{2} \log|\cos^2 x - 1/2| + c$$

on any interval I that does not contain points of the form $x = \pi/2 + k\pi$ or of the form $x = \pi/4 + k\pi/2$ with k an integer.

(d) With the substitution $y = \sqrt{x}$, $x, y > 0$, we have $x = y^2$, we write $dx = 2y \, dy$ and we obtain

$$\int \frac{x + \sqrt{x}}{2 + \sqrt{x}} dx = 2 \int \frac{y^3 + y^2}{y + 2} dy.$$

Integrating the rational function, we have

$$\begin{aligned} 2 \int \frac{y^3 + y^2}{y + 2} dy &= 2 \int (y^2 - y + 2) dy - 8 \int \frac{1}{y + 2} dy \\ &= \frac{2}{3} y^3 - y^2 + 4y - 8 \log(y + 2) + c. \end{aligned}$$

Substituting back $y = \sqrt{x}$, we conclude

$$\int \frac{x + \sqrt{x}}{2 + \sqrt{x}} dx = \frac{2}{3} x \sqrt{x} - x + 4\sqrt{x} - 8 \log(\sqrt{x} + 2) + c$$

on any interval $I \subset (0, +\infty)$.

(e) Using the rationalizing substitution $y = \tan \frac{x}{2}$, $dx = \frac{2}{1+y^2} dy$ and taking into account that $\cos x = \frac{1-y^2}{1+y^2}$, we have

$$\int \frac{1}{\cos x} dx = -2 \int \frac{1}{y^2 - 1} dy.$$

Using partial fractions, we then have

$$-2 \int \frac{1}{y^2 - 1} dy = \int \left(\frac{1}{y + 1} - \frac{1}{y - 1} \right) dy = \log \left| \frac{y + 1}{y - 1} \right| + c.$$

Going back to $y = \tan \frac{x}{2}$, we conclude

$$\int \frac{1}{\cos x} dx = \log \left| \frac{\tan(x/2) + 1}{\tan(x/2) - 1} \right| + c$$

on any interval I that does not contain points of the form $x = \pi/2 + k\pi$ with k an integer.

(f) Using the rationalizing substitution $x = 2 \sin y$, $-\pi/2 < y < \pi/2$, $-2 < x < 2$, $dx = 2 \cos y \, dy$, and taking into account also that $\cos y > 0$, we obtain

$$\int \sqrt{4-x^2} \, dx = 4 \int \cos^2 y \, dy.$$

From $4 \cos^2 y = 2 + 2 \cos(2y)$, we then have

$$4 \int \cos^2 y \, dy = 2y + \sin(2y) = 2y + 2 \sin y \cos y + c.$$

Now using $\sin y = x/2$, $y = \arcsin(x/2)$, $\cos y = \sqrt{1 - \sin^2 y} = \frac{1}{2} \sqrt{4-x^2}$, we conclude

$$\int \sqrt{4-x^2} \, dx = 2 \arcsin(x/2) + (x/2) \sqrt{4-x^2} + c$$

on any interval $I \subset (-2, 2)$.

Exercise 6

Compute the following limits

$$(a) \lim_{x \rightarrow 0^+} \frac{1}{x^3} \int_0^{x^2} \log(1 + \sqrt{t}) \, dt, \quad (b) \lim_{x \rightarrow 0^+} \frac{1}{x^4} \int_0^{x^2} (1 - \cos \sqrt{t}) \, dt$$

$$(c) \lim_{k \rightarrow +\infty} \left(2 \int_4^{4e} \frac{\log(kx)}{x} \, dx - \log(k(k+1)) \right), \quad (d) \lim_{k \rightarrow +\infty} \left(\int_1^3 \log(k(x+2)) \, dx - 3 \log(k+1) \right)$$

(a) Setting $f(x) = \int_0^{x^2} \log(1 + \sqrt{t}) dt$, the limit takes the indeterminate form $\frac{0}{0}$ given by $\lim_{x \rightarrow 0^+} \frac{f(x)}{x^3}$. The hypotheses of L'Hôpital's rule are satisfied, so we can move on to computing the limit

$$\lim_{x \rightarrow 0^+} \frac{f'(x)}{3x^2}.$$

We compute the derivative $f'(x)$ using the Fundamental theorem of calculus and the chain rule (derivative of a composite function):

$$f'(x) = 2x \log(1 + \sqrt{x^2}) = 2x \log(1 + x), \quad x > 0.$$

In conclusion

$$\begin{aligned} \lim_{x \rightarrow 0^+} \frac{1}{x^3} \int_0^{x^2} \log(1 + \sqrt{t}) dt &= \lim_{x \rightarrow 0^+} \frac{2x \log(1 + x)}{3x^2} \\ &= \frac{2}{3} \lim_{x \rightarrow 0^+} \frac{\log(1 + x)}{x} = \frac{2}{3}. \end{aligned}$$

(b) Setting $f(x) = \int_0^{x^2} (1 - \cos \sqrt{t}) dt$, the limit takes the indeterminate form $\frac{0}{0}$ given by $\lim_{x \rightarrow 0^+} \frac{f(x)}{x^4}$. The hypotheses of L'Hôpital's rule are satisfied, so we can move on to computing the limit

$$\lim_{x \rightarrow 0^+} \frac{f'(x)}{4x^3}.$$

We compute the derivative $f'(x)$ using the Fundamental theorem of calculus and the chain rule (derivative of a composite function):

$$f'(x) = 2x(1 - \cos \sqrt{x^2}) = 2x(1 - \cos x), \quad x > 0.$$

In conclusion

$$\begin{aligned} \lim_{x \rightarrow 0^+} \frac{1}{x^4} \int_0^{x^2} (1 - \cos \sqrt{t}) dt &= \lim_{x \rightarrow 0^+} \frac{2x(1 - \cos x)}{4x^3} \\ &= \frac{1}{2} \lim_{x \rightarrow 0^+} \frac{1 - \cos x}{x^2} = \frac{1}{4}. \end{aligned}$$

(c) We compute

$$\begin{aligned}
 2 \int_4^{4e} \frac{\log(kx)}{x} dx &= [\log^2(kx)]_4^{4e} = \log^2(4ke) - \log^2(4k) \\
 &= (\log(4ke) - \log(4k))(\log(4ke) + \log(4k)) \\
 &= \log \frac{4ke}{4k} \log(16k^2e) = \log(16k^2e).
 \end{aligned}$$

It follows that

$$\begin{aligned}
 \lim_{k \rightarrow +\infty} \left(2 \int_4^{4e} \frac{\log(kx)}{x} dx - \log(k(k+1)) \right) &= \lim_{k \rightarrow +\infty} \log(16k^2e) - \log(k(k+1)) \\
 &= \lim_{k \rightarrow +\infty} \log \frac{16k^2e}{k^2+k} = \log(16e) = 1 + \log 16.
 \end{aligned}$$

(d) We compute

$$\begin{aligned}
 \int_1^3 \log(k(x+2)) dx &= [x \log(kx+2k)]_1^3 - \int_1^3 \frac{x}{x+2} dx \\
 &= 3 \log(5k) - \log(3k) - \int_1^3 \left(1 - \frac{2}{x+2} \right) dx \\
 &= 3 \log(5k) - \log(3k) - 2 + 2 \log \frac{5}{3}.
 \end{aligned}$$

It follows that

$$\begin{aligned}
 \lim_{k \rightarrow +\infty} \left(\int_1^3 \log(k(x+2)) dx - 3 \log(k+1) \right) \\
 &= \lim_{k \rightarrow +\infty} 3 \log(5k) - \log(3k) - 2 + 2 \log \frac{5}{3} - 3 \log(k+1) \\
 &= \lim_{k \rightarrow +\infty} 3 \log \frac{5k}{k+1} - \log(3k) - 2 + 2 \log \frac{5}{3} = -\infty.
 \end{aligned}$$